

Gamma Approximation Validation for Active STAR-RIS Assisted UAV Networks With NOMA

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I. REASONABILITY OF GAMMA APPROXIMATION

In this supplementary note, we discuss the accuracy of the moment-matching based Gamma approximation used in the analytical derivation of the main paper. Specifically, the moment-matching method is employed to approximate the distribution of the cascaded channel gain as a Gamma distribution, which enables tractable analytical derivations for the SINR distributions and the corresponding downlink performance metrics.

This approximation method has been widely adopted in the performance analysis of RIS-assisted communication systems due to its analytical simplicity and good accuracy. In fact, a large number of existing studies have used the moment-matching based Gamma approximation to model the channel gain or SINR-related random variables in (STAR) RIS-assisted and NOMA-assisted communication systems, such as **Ref.[1]**, [7], [9], [33] and [37]. From the corresponding conclusions and numerical validations reported in these works, it can be seen that the approximation accuracy is generally high, and the analytical results obtained by this method usually match the simulation results well. Therefore, these existing studies indicate that the moment-matching based Gamma approximation is a mature and widely adopted analytical tool for RIS/STAR-RIS assisted communication systems.

On the other hand, from a mathematical point of view, applying the moment-matching method is also reasonable. Firstly, the cascaded channel gain (i.e., $|\chi_\kappa|^2$) in our work is a non-negative random variable, while the Gamma distribution is defined on the positive real axis, which makes it suitable for characterizing such quantities. Secondly, the moment-matching method preserves the first-order and second-order statistics of the original random variable, namely the mean and variance. These low-order moments are important for evaluating the considered performance metrics, including coverage probability *CP*, delay outage rate *DOR*, ergodic capacity *EC* and energy efficiency *EE*. In addition, the cascaded channel gain is formed by the combination of multiple positive random components, whose exact distribution is generally difficult to obtain in a closed form but usually exhibits a positively supported and right-skewed shape. Therefore, approximating the original variable as a Gamma distribution through the moment-matching method is theoretically feasible.

II. KOLMOGOROV-SMIRNOV BASED ERROR EVALUATION

To further evaluate the approximation accuracy, we compare the empirical distribution obtained from Monte Carlo simulations with the corresponding fitted Gamma distribution. The

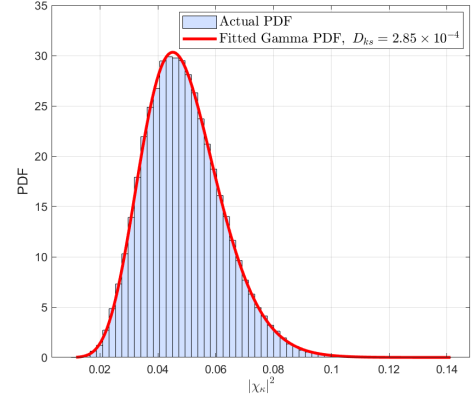


Fig. 1. Comparison of actual and fitted Gamma PDF.

Kolmogorov-Smirnov (KS) deviation is employed to measure the maximum difference between the approximated distribution and the actual empirical distribution **Ref.[2]**, which is defined as

$$\mathcal{D}_{KS} = \max |F_{\text{approx}}(x) - F_{\text{actual}}(x)| \quad (1)$$

where $F_{\text{approx}}(x)$ denotes the CDF of the fitted Gamma distribution and $F_{\text{actual}}(x)$ represents the empirical CDF obtained from Monte Carlo simulations.

As shown in fig. 1, the actual PDF of $|\chi_\kappa|^2$ is generated from 10^5 independent channel realizations. The fitted Gamma PDF closely matches the simulation histogram over the main range of the random variable, which indicates that the Gamma distribution provides a good approximation of the cascaded channel gain distribution under the considered simulation settings. Furthermore, the calculated $\mathcal{D}_{KS}$ is on the order of 10^{-4} , which further verifies the accuracy of the moment-matching based Gamma approximation in the considered cases.

It should be noted that this validation is conducted under the channel and mobility settings considered in the main paper. For highly dynamic environments with stronger temporal correlation, more complicated mobility patterns, or different fading statistics, the approximation accuracy may need to be re-evaluated. Nevertheless, under the considered simulation settings, the moment-matching based Gamma approximation provides a close match between the analytical and empirical distributions, and thus supports the tractability and accuracy of the proposed performance analysis.

Ref.[1] and **Ref.[2]** are listed as follows:

Ref.[1]: T. Wang, M. -A. Badiu, G. Chen and J. P. Coon, “Outage Probability Analysis of STAR-RIS Assisted NOMA Network With Correlated Channels,” *IEEE Commun. Lett.*, vol. 26, no. 8, pp. 1774-1778, Aug. 2022.

Ref.[2]: Z. Cui, K. Guan, J. Zhang and Z. Zhong, “SNR Coverage Probability Analysis of RIS-Aided Communication Systems,” *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3914–3919, Apr. 2021.